

Envision, Attend, Then Respond: Counterfactual Hallucination Mitigation in Large Vision-Language Models

Yuxuan Liang*, Fan Shi*, Rui Zhu, Xu Li, Xiaolei Chen, Zhe Liu, Bin Li†, Xiangyang Xue†

College of Computer Science and Artificial Intelligence, Fudan University

{yqliang25, fshi22, rzhu24, xu_li23, chenxl23, zheliu24}@m., {libin, xyxue}@fudan.edu.cn

Abstract

Large Vision-Language Models (LVLMs) often hallucinate when visual evidence conflicts with world knowledge, i.e., in counterfactual scenarios. We propose **Envision-Attend-Respond (EnAR)**, a training-free framework that leverages visual priors to steer the model’s attention toward counterfactual elements in the image. The **Envision** stage constructs a visual impression by invoking a diffusion prior to perform latent perturbations, yielding a prior-consistent counterpart of the input image. The **Attend** stage processes the original image and its visual impression through the LVLM’s vision encoder to localize counterfactual elements, forming a corresponding padded input. The **Respond** stage performs contrastive decoding between the original and padded inputs to suppress bias and enhance visual understanding. Empirically, EnAR consistently mitigates hallucinations and improves response fidelity, achieving a 10.82% gain on VLMBias and an average 6.9% improvement on POPE, demonstrating robustness across both counterfactual and general hallucination settings. Moreover, the framework remains effective across heterogeneous LVLM architectures, offering a new perspective for hallucination governance in multimodal reasoning. The code is available at <https://github.com/Lyxxx1211/CVPR2026-EnAR>.

1. Introduction

Large Vision-Language Models (LVLMs) have recently demonstrated remarkable progress in unifying visual perception with linguistic reasoning [2, 5, 9, 15, 42, 50, 51]. By aligning pre-trained visual encoders with Large Language Models (LLMs) [4, 13, 21, 38, 43, 47, 61, 67] via multimodal adapters, LVLMs process visual and textual inputs within a shared semantic space and deliver strong results on various vision-language tasks, including image captioning, visual question answering, and multimodal dialogue.

*Equal contribution

†Corresponding author

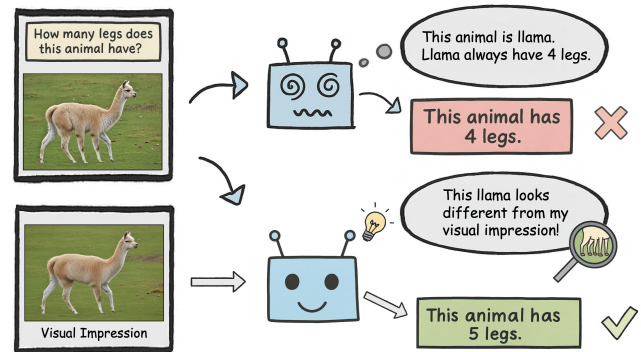


Figure 1. **Illustration of the Envision-Attend-Respond framework.** By comparing the visual impression with the original image to locate counterfactual elements, the model is guided toward producing the correct response.

Despite these advances, LVLMs remain vulnerable to hallucinations, where generated text is inconsistent with the actual visual content [28, 44, 68]. Such behavior severely undermines model reliability in safety-critical applications such as autonomous driving [12] and medical image analysis [40]. Among all hallucinations, a particular type is counterfactual hallucination, which arises when the input image contradicts commonsense world knowledge [7, 53, 68]. As shown in Figure 1, when an LVLM is presented with a picture of a five-legged llama and asked about the number of legs, it incorrectly answers that the llama has four. Since the LLM component accumulates extensive world knowledge during web-scale pretraining, linguistic priors can dominate text generation over the visual evidence. Recent studies show that even the most advanced LVLMs suffer substantial performance drops in counterfactual settings [68], far below human-level understanding. These observations highlight the need to explicitly address counterfactual hallucination and ensure that LVLMs ground their reasoning in perception rather than priors.

Various strategies have been proposed to mitigate hallucinations in LVLMs. At the data level [10, 49, 78, 79], new visual instruction tuning datasets have been created

to strengthen the alignment between textual elements and image content, thereby improving the model’s ability to ground language in visual evidence. At the encoding level [8, 54, 66, 72], approaches that integrate multiple visual encoders leverage their complementary strengths, enhancing the model’s ability to extract richer visual features. At the training level [16, 29, 75, 79, 82], reinforcement learning-based techniques have been introduced to encourage models to better align text generation with observed visual reality. At the decoding level [3, 23, 31, 39, 74, 80], methods have been developed to generate logit pairs for contrastive decoding, reducing the likelihood of sampling hallucinated tokens during text generation. While these methods show promise in reducing hallucinations in general, they mainly target broad hallucination phenomena and do not specifically address counterfactual cases. As a result, a specialized approach for mitigating counterfactual hallucinations is valuable, as current solutions fail to effectively handle situations where visual evidence contradicts prior knowledge.

Humans develop visual impressions through long-term exposure to visual experiences [18, 24, 35, 64, 65], building upon prior knowledge accumulated from repeated interactions with the environment. These impressions encompass not only explicit features like appearance, color, and shape, but also implicit information regarding objects’ behavior, use, and contextual associations. For instance, when encountering a common four-legged animal such as a llama, our visual impression strongly informs us that it should have four legs, forming a robust expectation based on prior knowledge [14, 65]. However, when faced with an anomalous scene, like a five-legged llama, a cognitive conflict arises between the visual impression and the actual scene [19, 25, 32]. This discrepancy triggers a shift in attention toward the counterfactual element (i.e., the fifth leg of the llama). This contrastive perception mechanism helps us detect inconsistencies in the visual input and reassess the actual content. Rather than passively accepting the visual information, we engage in counterfactual reasoning, using our visual impressions to adjust our understanding of the world. This ability is crucial for making swift, accurate judgments in complex environments [36, 37, 62], allowing us to focus on counterfactual elements when faced with unexpected scenarios, ultimately reducing erroneous conclusions and ensuring flexibility in decision-making.

Inspired by this process, we propose a novel framework **Envision-Attend-Respond (EnAR)** for mitigating counterfactual hallucinations in LVLMs, as illustrated in Figure 1. In the *Envision* stage, we utilize a pre-trained diffusion model [30, 55, 57, 63] to generate an edited version of the input image as the visual impression. As the diffusion model is pre-trained on vast amounts of visual data, this impression represents an idealized version of the scene based on prior knowledge, while also providing a pixel-wise

uncertainty map that quantifies areas of uncertainty during the generation process. In the *Attend* stage, we draw on the visual attention mechanism of the LVLM to compare the attention distributions between the generated visual impression and the actual input image. This comparison helps to localize the counterfactual elements in the image, similar to how humans focus on areas of inconsistency. Once these regions are identified, we replace the counterfactual tokens with a pad token, creating a second visual input stream in which the inconsistent areas are masked out. In the *Respond* stage, we allow the model to focus on the true visual content by contrastively decoding both input streams, which reduces biases and ensures the model generates responses grounded in visual evidence. Our main contributions are summarized as follows:

- We introduce a method to construct *visual impressions* of an image by harnessing the diffusion model’s internalized real-world visual prior, along with a pixel-wise uncertainty map, which jointly benefit precise localization of counterfactual elements.
- We propose the EnAR framework, a training-free framework that mitigates counterfactual hallucinations in LVLMs by guiding the model’s attention and decoding toward perception-grounded evidence.
- We validate EnAR across three distinct LVLM families, including InternVL-3.5 [71], Qwen2.5VL [6], and LLaVA-v1.5 [51], on five hallucination benchmarks, demonstrating consistent improvements, including a **10.82%** gain on VLMBias [68] and an average **6.9%** improvement on POPE [44].

2. Related Work

2.1. Large Vision-Language Models

Large Vision-Language Models (LVLMs) combines image-text contrastively pre-trained vision encoders with Large Language Models, enabling joint interpretation of images and text. Early LVLMs such as InsturctBlip [15] and LLaVA-v1.5 [51] connected frozen CLIP-based [20, 56] visual transformers to LLMs via lightweight adapters, which allowed open-ended visual question answering. However, these models often operate at limited input resolutions and require downsampling high-resolution images, hindering their ability to capture fine-grained details. To address this limitation, subsequent LVLMs [11, 45, 46, 52, 69, 70, 76, 77, 83] employed dynamic cropping of high-resolution images or advanced training strategies to preserve visual details. Notably, recent models like InternVL3.5 [71] and Qwen2.5VL [6] introduce novel frameworks (e.g. cascade reinforcement learning and dynamic resolution encoding) that boost multimodal reasoning performance, closing the gap between open-source models and proprietary models like GPT-4V [1]. However, alongside these improvements,

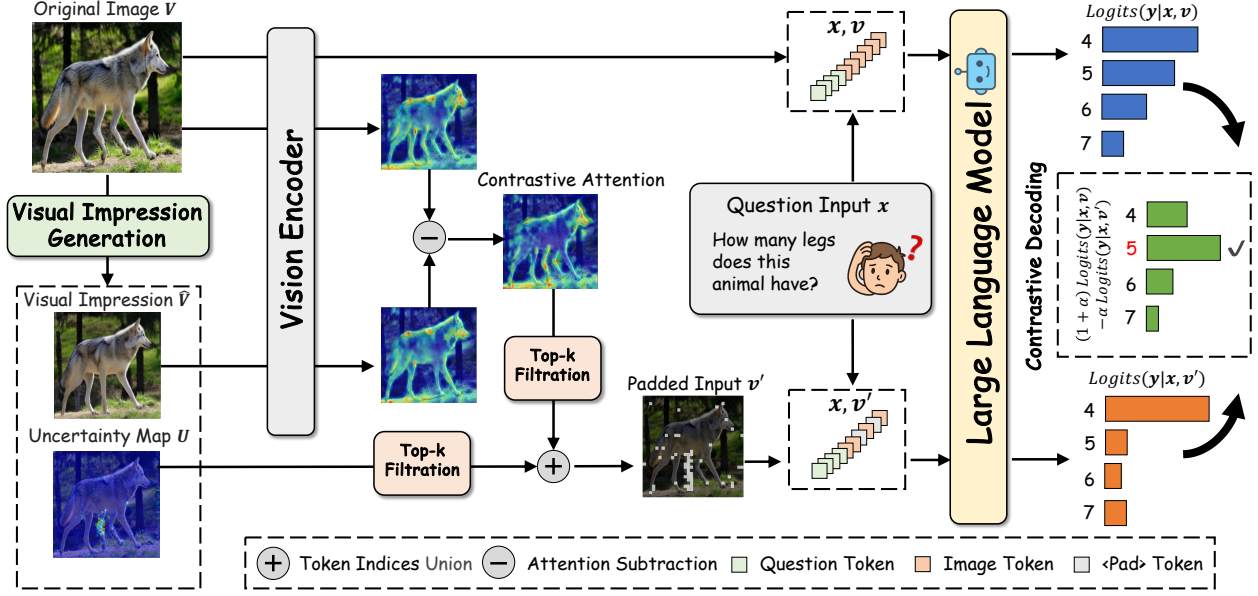


Figure 2. **Overview of the EnAR framework.** EnAR first generates a visual impression with a corresponding pixel-wise uncertainty map. Next, EnAR feeds both the original and impression images into the LVLM’s vision encoder to obtain contrastive attention, pads counterfactual tokens by combining attention differences with the uncertainty map. Finally, EnAR performs contrastive decoding over the original and padded inputs to suppress counterfactual hallucinations and produce the final response.

LVLMs still tend to produce hallucinations [28, 44] which remains a key limitation on their reliability.

2.2. Hallucination Mitigation in LVLMs

Many studies have analyzed the causes of hallucinations in LVLMs and provided benchmarks to assess the severity of hallucinations [7, 26, 28, 44, 53, 66, 68]. To mitigate hallucinations, researchers have proposed different strategies at various stages of the vision-language pipeline. At the data, encoding, and training levels, numerous works [8, 54, 66, 72, 75, 78, 79, 82] have introduced effective strategies to mitigate general hallucination phenomena.

While the methods above usually require additional model training, a number of training-free methods have been proposed to curb hallucinations at the LVLM decoding stage. VCD [39] adds Gaussian noise to the input image, then contrasts the model’s outputs between the original and the noisy images. Similarly, RITUAL [74] feeds randomly generated copies of the image (e.g. rotated or cropped versions) and contrasts with the original one during decoding. M3ID [23] contrasts the logits between the original input and text-only input. DeGF [80] uses the model’s own initial description to synthesize an image, which is then fed back into the LVLM to verify whether the described content is consistent with visual reality. Additionally, AGLA [3] addresses hallucination by masking out prompt-irrelevant regions of the image. All these studies have achieved progress in mitigating hallucinations in LVLMs, effectively address-

ing general hallucination scenarios. Nevertheless, these methods remain ineffective when confronted with counterfactual hallucinations and struggle to localize the counterfactual elements within the image. To bridge this gap, we propose a training-free framework that envisions a visual impression of the input image, attends to counterfactual elements, and finally performs contrastive decoding to generate more accurate responses.

3. Method

As shown in Figure 2, EnAR contains three stages: Envision, Attend, and Respond. In the *Envision* stage, we derive a visual impression and a pixel-wise uncertainty map. In *Attend*, we jointly use attention and uncertainty maps to locate counterfactual elements and construct a padded visual input. In *Respond*, we apply contrastive decoding between the original and the padded inputs to suppress counterfactual hallucinations. The details of our pipeline are introduced in the following sections.

3.1. Envision: Visual Impression Generation

The first stage aims to generate a visual impression of the input image V , which is illustrated in Figure 3. When the input contains counterfactual elements (e.g., a wolf with five legs in Figure 2), humans can form an intuition of *what it should have been*. To formalize this intuition, we introduce the visual impression $\hat{V}$, which represents an approximation of V under the real-world visual prior. The vi-

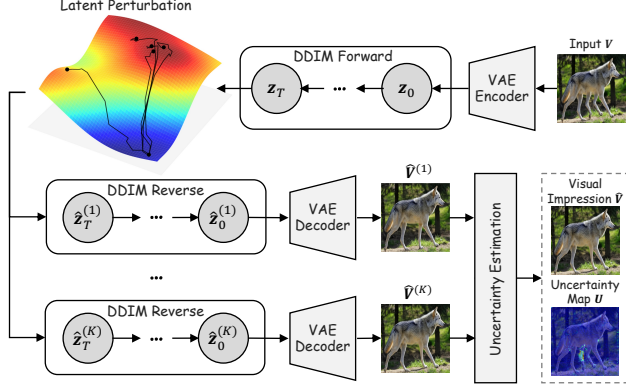


Figure 3. Illustration of the *Envision* stage in EnAR.

sual impression is designed to preserve the normal regions of the input while transforming the implausible regions toward patterns that are more likely to appear in the prior. Deep generative models [27, 30, 34, 59, 60] encode large-scale visual priors by parameterizing the data distribution $p(\mathbf{V})$, among which diffusion models are suited to our setting. First, they are usually trained on large-scale image datasets, providing a substantially richer and more realistic visual prior. Second, the diffusion formulation offers an explicit gradient field $\nabla_{\mathbf{V}} \log p(\mathbf{V})$ that can effectively guide the input toward regions of higher visual plausibility.

Background and Preliminaries. We can get the evolution trajectory of $\mathbf{V}$ in the latent space of a diffusion model [30, 57, 63]. The input and its latent representation are connected through a VAE, that is, $\mathbf{z}_0 = \text{Encoder}(\mathbf{V})$ and $\mathbf{V} = \text{Decoder}(\mathbf{z}_0)$. The diffusion model gradually adds Gaussian noise to the latent representation, producing a sequence of intermediate latents indexed by $t \in \{1, \dots, T_{\max}\}$. Given the Gaussian variance β_t at each diffusion step, we obtain the following expression of the forward process.

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (1)$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ and $\alpha_t = 1 - \beta_t$. A neural network ε_θ is trained to predict the noise injected at each step to recover $\mathbf{z}_0$ in the reverse process, which can be written as

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(\mathbf{z}_t, t) \right) + \tilde{\beta}_t \boldsymbol{\varepsilon}, \quad (2)$$

where $\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Since Equations 1 and 2 are stochastic, the DDPM sampling process is difficult to invert and is therefore incompatible with applications that require precise reconstruction or controlled refinement of a given input latent. Therefore, we adopt the deterministic DDIM sampler [63], which provides deterministic forward and reverse updates. This allows us to transform $\mathbf{z}_0$ to the diffusion step $\mathbf{z}_T$ for modification. We next introduce a perturbation mechanism that locally modifies the intermediate latent $\mathbf{z}_0$ under the guidance of the gradient field.

Algorithm 1 Visual Impression Generation

Require: input image $\mathbf{V}$, inversion step T , number of iterations M , number of samples K

Ensure: visual impression $\hat{\mathbf{V}}$, uncertainty map $\mathbf{U}$

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1:  $\mathbf{z}_T = \text{DDIMForward}(\text{Encoder}(\mathbf{V}), T)$ 
2: for  $k = 1$  to  $K$  do
3:    $\hat{\mathbf{z}}_T^{(k)} \leftarrow \mathbf{z}_T$ 
4:   for  $m = 1$  to  $M$  do
5:      $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
6:      $\hat{\mathbf{z}}_T^{(k)} \leftarrow \hat{\mathbf{z}}_T^{(k)} - \eta \cdot \frac{\varepsilon_\theta(\hat{\mathbf{z}}_T^{(k)}, T)}{\sqrt{1 - \bar{\alpha}_T}} + \sqrt{2\eta\tau} \cdot \boldsymbol{\xi}$ 
7:    $\hat{\mathbf{V}}^{(k)} = \text{Decoder}(\text{DDIMReverse}(\hat{\mathbf{z}}_T^{(k)}, T))$ 
8:  $\hat{\mathbf{V}}, \mathbf{U} = \text{UncertaintyEstimation}(\{\hat{\mathbf{V}}^{(k)}\}_{k=1}^K)$ 
9: return  $\hat{\mathbf{V}}, \mathbf{U}$ 

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Latent Perturbation. We perturb the intermediate latent representation $\mathbf{z}_T$ to generate the visual impression. Since the objective is to modify only the counterfactual elements while preserving the normal structures, it is essential to determine an appropriate modification direction in the latent space. EnAR utilizes the gradient field $\mathbf{G} = \nabla_{\mathbf{z}_T} \log p(\mathbf{z}_T)$, which points toward regions of higher likelihood under the real-world visual prior. This gradient field can be estimated using the Tweedie estimator [22]:

$$\mathbf{G} = \frac{\alpha_T \mathbb{E}[\mathbf{z}_0 | \mathbf{z}_T] - \mathbf{z}_T}{1 - \bar{\alpha}_T} \approx -\frac{\varepsilon_\theta(\mathbf{z}_T, T)}{\sqrt{1 - \bar{\alpha}_T}}. \quad (3)$$

The detailed derivation is provided in Appendix A. To refine $\mathbf{z}_T$ toward regions of higher likelihood, we employ annealed Langevin dynamics sampling [73]:

$$\mathbf{z}_T \leftarrow \mathbf{z}_T + \eta \cdot \mathbf{G} + \sqrt{2\eta\tau} \cdot \boldsymbol{\xi}, \quad (4)$$

where $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the step size η is annealed from 10^{-2} to 10^{-4} , and the temperature is $\tau = 0.1$. We apply perturbation at $T = 30$ (total $T_{\max} = 50$ steps) and perform $M = 10$ steps of Langevin sampling, which strikes a balance to preserve the global structure of the input while modifying counterfactual elements [58]. After latent perturbation, the deterministic DDIM reverse process is applied to obtain $\hat{\mathbf{z}}_0$, which is subsequently decoded to produce the visual impression.

Uncertainty Estimation. To estimate the uncertainty of generating the visual impression, EnAR performs K perturbations on $\mathbf{z}_T$ to produce K visual impressions $\{\hat{\mathbf{V}}^{(k)}\}_{k=1}^K$. EnAR computes the variance for each location (i, j) across the impressions to construct an uncertainty map $\mathbf{U}$ where $U_{i,j} = \text{Var}(\{\hat{\mathbf{V}}_{i,j}^{(k)}\}_{k=1}^K)$. To obtain a representative visual impression, EnAR measures the deviation of each generated impression from the input and selects the one that deviates the most. The final visual impression is defined as

Table 1. **Performance comparison on VLMBias across different categories.** We report accuracy (%) for seven hallucination categories and overall performance across three baseline models. Values in parentheses indicate performance changes relative to Regular. Blue denotes improvement, gray denotes degradation, and the best-performing results are highlighted in bold.

Model	Method	Animals	Chess Pieces	Flags	Game Boards	Logos	Optical Illusion	Patterned Grid	Overall
InternVL3.5-8B [71]	Regular	0.00	0.00	25.00	8.93	5.07	48.48	21.43	19.83
	+VCD[39]	14.29 (14.29)	0.00 (0.00)	28.75 (3.75)	14.29 (5.36)	7.86 (2.79)	51.51 (3.03)	15.18 (6.25)	23.76 (3.93)
	+M3ID[23]	0.00 (0.00)	0.00 (0.00)	25.00 (0.00)	8.93 (0.00)	5.00 (0.07)	49.62 (1.14)	21.43 (0.00)	20.11 (0.28)
	+RITUAL[74]	0.00 (0.00)	0.00 (0.00)	22.50 (2.50)	7.14 (1.79)	2.86 (2.21)	49.24 (0.76)	16.96 (4.47)	18.82 (1.01)
	+DeGF[80]	5.49 (5.49)	0.00 (0.00)	25.00 (0.00)	10.71 (1.78)	5.71 (0.64)	49.62 (1.14)	20.54 (0.89)	21.29 (1.46)
	+AGLA[3]	0.00 (0.00)	0.00 (0.00)	27.50 (2.50)	8.93 (0.00)	3.57 (1.50)	49.62 (1.14)	20.54 (0.89)	20.00 (0.17)
	+EnAR	25.82 (25.82)	0.00 (0.00)	30.00 (5.00)	16.07 (7.14)	34.78 (29.71)	52.27 (3.79)	22.32 (0.89)	31.36 (11.53)
Qwen2.5VL-7B [6]	Regular	0.00	0.00	20.00	3.75	12.32	60.23	14.29	22.63
	+VCD[39]	9.34 (9.34)	0.00 (0.00)	23.75 (3.75)	7.14 (3.39)	17.39 (5.07)	56.44 (3.79)	15.18 (0.89)	24.78 (2.15)
	+M3ID[23]	0.00 (0.00)	0.00 (0.00)	20.00 (0.00)	3.75 (0.00)	12.32 (0.00)	60.23 (0.00)	14.29 (0.00)	22.63 (0.00)
	+RITUAL[74]	0.00 (0.00)	0.00 (0.00)	20.00 (0.00)	3.57 (0.18)	9.42 (2.90)	58.71 (1.52)	11.61 (2.68)	21.44 (1.19)
	+DeGF[80]	3.85 (3.85)	0.00 (0.00)	20.00 (0.00)	3.57 (0.18)	10.14 (2.18)	59.85 (0.38)	12.50 (1.79)	22.73 (0.10)
	+AGLA[3]	0.00 (0.00)	0.00 (0.00)	18.75 (1.25)	3.57 (0.18)	9.42 (2.90)	59.09 (1.14)	14.29 (0.00)	21.77 (0.86)
	+EnAR	18.68 (18.68)	0.00 (0.00)	23.75 (3.75)	7.14 (3.39)	16.67 (4.35)	61.74 (1.51)	15.18 (0.89)	28.02 (5.39)
LLaVA-v1.5-7B [51]	Regular	0.00	0.00	8.75	10.71	8.70	50.0	0.00	16.92
	+VCD[39]	1.09 (1.09)	0.00 (0.00)	7.50 (1.25)	5.36 (5.35)	13.77 (5.07)	48.11 (1.89)	18.75 (18.75)	19.18 (2.26)
	+M3ID[23]	0.00 (0.00)	0.00 (0.00)	8.75 (0.00)	10.71 (0.00)	8.70 (0.00)	50.00 (0.00)	0.00 (0.00)	16.92 (0.00)
	+RITUAL[74]	0.00 (0.00)	0.00 (0.00)	7.50 (1.25)	7.14 (3.57)	8.70 (0.00)	50.38 (0.38)	0.00 (0.00)	16.70 (0.22)
	+DeGF[80]	0.55 (0.55)	0.00 (0.00)	7.50 (1.25)	1.79 (8.92)	9.42 (0.72)	51.52 (1.52)	16.96 (16.96)	18.97 (2.05)
	+AGLA[3]	0.00 (0.00)	0.00 (0.00)	10.00 (1.25)	7.14 (3.57)	9.42 (0.72)	51.14 (1.14)	1.79 (1.79)	17.56 (0.64)
	+EnAR	3.85 (3.85)	0.00 (0.00)	12.50 (3.75)	10.71 (0.00)	13.04 (4.34)	53.03 (3.03)	21.43 (21.43)	22.20 (5.28)

$\hat{\mathbf{V}} = \hat{\mathbf{V}}^{(k^*)}$ where $k^* = \arg \max_k \|\hat{\mathbf{V}} - \hat{\mathbf{V}}^{(k)}\|_2^2$. The full generation process is provided in Algorithm 1.

3.2. Attend: Counterfactual Element Localization

After obtaining the visual impression $\hat{\mathbf{V}}$, we next aim to locate the counterfactual elements which conflict with the world knowledge. To expose such elements, we analyze the attention distributions derived from the original image and its impression counterpart, leveraging the inherent self-attention mechanism of the LVLM’s vision encoder.

Specifically, from the L -th transformer layer of the vision encoder, we extract the attention weights assigned by the $\langle cls \rangle$ token to patch tokens to measure how the model attends to different spatial locations. For models whose vision encoders do not contain a $\langle cls \rangle$ token (e.g., Qwen-VL series), we obtain the attention distribution by summing each token’s incoming attention within the layer and then average it across heads. Given the original image $\mathbf{V}$ and its impression $\hat{\mathbf{V}}$, their corresponding attention maps in vision encoder are denoted as $\text{Attn}^{(L)}(\mathbf{V})$ and $\text{Attn}^{(L)}(\hat{\mathbf{V}})$. We then compute an element-wise absolute difference to reveal attention shift induced by counterfactual content:

$$\Delta \mathbf{A} = |\text{Attn}^{(L)}(\mathbf{V}) - \text{Attn}^{(L)}(\hat{\mathbf{V}})|. \quad (5)$$

Here, $\Delta \mathbf{A}$ represents a contrastive attention map, where larger values indicate the tokens are potentially associated with the counterfactual elements.

To identify the most suspicious locations, we select the top- $K\%$ token indices with the highest $\Delta \mathbf{A}$ scores and form an initial index set $\mathcal{H}_{\text{attn}}$. In parallel, we leverage the pixel-wise uncertainty map $\mathcal{U}$: the top 5% most uncertain posi-

tions are mapped to token indices to form $\mathcal{H}_{\text{unc}}$. We then leverage the union of the two sets to obtain the final counterfactual token index set:

$$\mathcal{H} = \mathcal{H}_{\text{attn}} \cup \mathcal{H}_{\text{unc}}. \quad (6)$$

Jointly considering the information derived from attention and uncertainty maps can lead to a more comprehensive counterfactual localization.

Finally, given the original visual embedding sequence $\mathbf{v}$, we replace the counterfactual tokens indexed by $\mathcal{H}$ with the model’s built-in padding token $\langle pad \rangle$, forming a padded input $\mathbf{v}'$. This operation constructs a contrastive visual input that forms a precise, token-level difference relative to the original image while keeping the original branch $\mathbf{v}$ intact.

3.3. Respond: Contrastive Decoding

Given an embedded question query $\mathbf{x}$ and the two visual inputs $(\mathbf{v}, \mathbf{v}')$, the logits of the next token y are denoted as $p(y|\mathbf{x}, \mathbf{v})$ and $p(y|\mathbf{x}, \mathbf{v}')$. Following the formulation in VCD [39], we compute the contrastive decoding logits as:

$$p(y|\mathbf{x}, \mathbf{v}, \mathbf{v}') = (1 + \alpha) p(y|\mathbf{x}, \mathbf{v}) - \alpha p(y|\mathbf{x}, \mathbf{v}'). \quad (7)$$

Here, α is a hyperparameter that controls the strength of contrastive suppression. This mechanism effectively highlights counterfactual elements, enhancing the reliability of the final response.

4. Experiments

4.1. Experimental Settings

Baseline Models. We evaluate EnAR on three distinct LVLMs: InternVL3.5-8B [71], Qwen2.5VL-7B [6], and

Table 2. **Performance comparison on PhD, WHOOPS, and HallusionBench benchmarks.** We selected the Counting subset of PhD, which best fits counterfactual scenarios, and further grouped the 26 categories in WHOOPS into three broader types: Social, Natural, and Symbolic. We report accuracy (%) for PhD and WHOOPS. We report Accuracy per Question (aAcc), Accuracy per Figure (fAcc), and Accuracy per Question Pair (qAcc) for three LVLm backbones.

Model	Method	PhD	WHOOPS				HallusionBench			
		Acc.	Social	Natural	Symbolic	Average	aAcc	fAcc	qAcc	Average
InternVL3.5-8B [71]	Regular	73.10	59.52	65.48	62.36	62.45	70.70	44.52	46.03	53.75
	+VCD[39]	72.75 (-0.35)	68.63 (+9.11)	67.26 (+1.78)	72.47 (+10.11)	69.45 (+7.00)	71.21 (+0.51)	45.10 (+0.58)	46.43 (+0.40)	54.25 (+0.50)
	+M3ID[23]	72.15 (-0.95)	69.44 (+9.92)	68.60 (+3.12)	74.72 (+12.36)	70.92 (+8.47)	71.19 (+0.49)	44.04 (-0.48)	46.47 (+0.44)	53.90 (+0.15)
	+RITUAL[74]	72.25 (-0.85)	69.17 (+9.65)	68.60 (+3.12)	74.72 (+12.36)	70.83 (+8.38)	70.22 (-0.48)	43.95 (-0.57)	44.18 (-1.85)	52.78 (-0.97)
	+DeGF[80]	73.00 (-0.10)	67.02 (+7.50)	66.15 (+0.67)	74.72 (+12.36)	69.30 (+6.85)	71.53 (+0.83)	44.69 (+0.17)	46.59 (+0.56)	54.27 (+0.52)
	+AGLA[3]	72.20 (-0.90)	70.51 (+10.99)	69.93 (+4.45)	75.84 (+13.48)	72.09 (+9.64)	70.73 (+0.03)	43.77 (-0.75)	44.96 (-1.07)	53.15 (-0.60)
	+EnAR	73.45 (+0.35)	71.85 (+12.33)	70.82 (+5.34)	79.78 (+17.42)	74.15 (+11.70)	71.93 (+1.23)	45.74 (+1.22)	46.83 (+0.80)	54.83 (+1.08)
Qwen2.5VL-7B [6]	Regular	71.35	47.88	55.76	44.38	49.34	71.73	47.58	47.68	55.66
	+VCD[39]	74.80 (+3.45)	60.59 (+12.71)	53.67 (-2.09)	46.63 (+2.25)	53.63 (+4.29)	70.17 (-1.56)	46.27 (-1.31)	46.16 (-1.52)	54.20 (-1.46)
	+M3ID[23]	74.35 (+3.00)	64.88 (+17.00)	56.35 (+0.59)	52.25 (+7.87)	57.83 (+8.49)	72.53 (+0.80)	47.37 (-0.21)	48.24 (+0.56)	56.05 (+0.39)
	+RITUAL[74]	73.95 (+2.60)	61.39 (+13.51)	55.01 (-0.75)	51.69 (+7.31)	56.03 (+6.69)	69.36 (-2.37)	40.18 (-7.40)	41.96 (-5.72)	50.50 (-5.16)
	+DeGF[80]	74.25 (+2.90)	65.42 (+17.54)	59.79 (+4.03)	52.25 (+7.87)	59.15 (+9.81)	71.94 (+0.21)	47.06 (-0.52)	48.33 (+0.65)	55.78 (+0.12)
	+AGLA[3]	73.30 (+1.95)	62.47 (+14.59)	54.79 (-0.97)	51.12 (+6.74)	56.13 (+6.79)	71.62 (-0.11)	46.28 (-1.30)	46.58 (-1.10)	54.83 (-0.83)
	+EnAR	75.62 (+4.27)	66.76 (+18.88)	60.58 (+4.82)	59.55 (+15.17)	62.30 (+12.96)	72.92 (+1.19)	48.03 (+0.45)	49.52 (+1.84)	56.82 (+1.16)
LLaVA-v1.5-7B [51]	Regular	59.10	6.70	6.90	6.18	6.52	35.28	18.42	10.32	21.34
	+VCD[39]	59.35 (+0.25)	6.70 (0.00)	7.57 (+0.67)	5.06 (-1.12)	6.44 (-0.08)	37.43 (+2.15)	18.68 (+0.26)	12.34 (+2.02)	22.82 (+1.48)
	+M3ID[23]	59.00 (-0.10)	6.70 (0.00)	6.90 (0.00)	6.18 (0.00)	6.59 (+0.07)	37.24 (+1.96)	18.17 (-0.25)	10.73 (+0.41)	22.05 (+0.71)
	+RITUAL[74]	59.25 (+0.15)	5.90 (-0.80)	6.90 (0.00)	6.18 (0.00)	6.33 (-0.19)	36.77 (+1.49)	18.34 (-0.08)	10.77 (+0.45)	21.96 (+0.62)
	+DeGF[80]	59.90 (+0.80)	4.29 (-2.41)	6.90 (0.00)	6.18 (0.00)	5.79 (-0.73)	37.14 (+1.86)	17.26 (-1.16)	12.53 (+2.21)	22.31 (+0.97)
	+AGLA[3]	60.30 (+1.20)	6.17 (-0.53)	7.57 (+0.67)	5.62 (-0.56)	6.45 (-0.07)	37.74 (+2.46)	17.87 (-0.55)	12.13 (+1.81)	22.58 (+1.24)
	+EnAR	60.55 (+1.45)	9.65 (+2.95)	11.36 (+4.46)	12.92 (+6.74)	11.31 (+4.79)	39.33 (+4.05)	20.03 (+1.61)	15.43 (+5.11)	24.93 (+3.59)

LLaVA-v1.5-7B [51]. These models serve as baselines to demonstrate the model-agnostic property of our framework. **Benchmarks.**

VLMBias: The VLMBias benchmark [68] is designed to test counterfactual visual hallucination in LVLms. It contains images in which well-known objects are subtly modified (e.g., adding an extra leg to a familiar animal or altering the stripes on an Adidas logo) and pairs them with objective questions (e.g., counting, yes/no identification) to reveal whether the model relies on memorized priors rather than actual visual evidence. The benchmark is explicitly high difficulty and probes models’ failures to detect non-canonical visual cues.

Others: We also evaluate EnAR on the counterfactual benchmarks PhD [53] and WHOOPS [7]. To assess language hallucination and visual illusion, we use HallusionBench [28]. For general hallucination assessment, we adopt POPE [44], a dataset constructed based on MSCOCO [48]. Details of these benchmarks are provided in Appendix B.1.

Implemental Details. We compare EnAR with different decoding level hallucination mitigation methods, including the model’s original decoding strategy (denoted as Regular), VCD [39], M3ID [23], RITUAL [74], DeGF [80], and AGLA [3]. For all methods, we adopt the best-performing configurations reported in their papers and reproduce the results by implementing their pipelines based on the released code and documented procedures, ensuring a fair comparison. Across all comparisons, EnAR consistently employs the 6th vision-encoder layer and a padding-token ratio of 10%. The visual impressions are generated using Stable

Diffusion v1.5 [57]. Our contrastive decoding implementation and hyperparameter setting follow the configuration of VCD. More details can be found in Appendix B.2.

4.2. Experimental Results

VLMBias. As shown in Table 1, EnAR consistently achieves the best performance across different LVLm backbones. On InternVL3.5-8B, overall accuracy rises from 19.83% to 31.36%, with large improvements on Logos and Animals. On QwenVL2.5-7B and LLaVA-1.5-7B, overall accuracy increases from 22.63% to 28.02% and from 16.92% to 22.20%, respectively. Compared with other methods, EnAR remains strongest at the overall level for each backbone (e.g., InternVL: 31.36% vs. 23.76% with VCD; LLaVA: 22.20% vs. 17.56% with AGLA). Notably, all of the methods obtain zero scores on Chess Pieces, which may stem from the baseline models’ limited capacity to handle such reasoning tasks, leaving little room for subsequent correction. The consistent gains on VLMBias highlight EnAR’s ability to redirect models toward visual evidence when world knowledge conflicts with the image. The improvements observed across heterogeneous architectures further underscore its model-agnostic robustness on this challenging benchmark.

PhD and WHOOPS. Table 2 presents the results on the PhD and WHOOPS benchmarks. EnAR consistently outperforms other methods across all LVLm backbones. For InternVL3.5-8B, EnAR achieves substantial gains on WHOOPS, elevating overall accuracy from 62.45% to 74.15%. Similarly, for Qwen2.5VL-7B, counting accuracy

Table 3. **Performance comparison on POPE benchmark across different methods.** We report Accuracy (Acc.), Precision (Prec.), Recall, and F1 scores for three LVLM backbones.

Setup	Method	LLaVA-v1.5-7B [51]				QwenVL2.5-7B [6]				InternVL3.5-8B [71]			
		Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Random	Regular	83.7	92.7	71.7	80.8	88.2	98.8	77.3	86.7	92.1	94.4	90.6	92.5
	VCD[39]	87.8 (4.1)	91.5 (1.2)	82.3 (10.6)	86.6 (5.8)	89.0 (0.8)	98.5 (0.3)	79.2 (1.9)	87.8 (1.1)	93.2 (1.1)	95.6 (1.2)	90.5 (0.1)	93.0 (0.5)
	M3ID[23]	88.5 (4.8)	92.1 (0.6)	82.1 (10.4)	86.8 (6.0)	87.9 (0.3)	99.0 (0.2)	76.4 (0.9)	86.2 (0.5)	93.2 (1.1)	95.4 (1.0)	90.8 (0.2)	93.0 (0.5)
	RITUAL[74]	88.5 (4.8)	92.8 (0.1)	80.6 (8.9)	86.2 (5.4)	88.4 (0.2)	98.6 (0.2)	77.8 (0.5)	87.0 (0.3)	93.0 (0.9)	95.7 (1.3)	90.1 (0.5)	92.8 (0.3)
	DeGF[80]	88.7 (5.0)	93.3 (0.6)	82.4 (10.7)	87.5 (6.7)	88.8 (0.6)	98.5 (0.3)	78.6 (1.3)	87.5 (0.8)	93.2 (1.1)	95.1 (0.7)	90.2 (0.4)	92.6 (0.1)
	AGLA[3]	88.4 (4.7)	94.4 (1.7)	82.0 (10.3)	87.7 (6.9)	88.2 (0.0)	99.1 (0.3)	78.0 (0.7)	87.3 (0.6)	93.1 (1.0)	95.6 (1.2)	90.1 (0.5)	92.8 (0.3)
	EnAR	89.6 (5.9)	95.5 (2.8)	83.0 (11.3)	88.9 (8.1)	89.9 (1.7)	98.5 (0.3)	81.0 (3.7)	88.9 (2.2)	93.7 (1.6)	95.9 (1.5)	91.0 (0.4)	93.4 (0.9)
Popular	Regular	81.5	88.3	71.7	79.1	87.5	96.8	77.3	86.0	87.1	84.6	90.6	87.5
	VCD[39]	85.8 (4.3)	86.4 (1.9)	82.2 (10.5)	84.2 (5.1)	88.2 (0.7)	96.6 (0.2)	79.2 (1.9)	87.0 (1.0)	87.1 (0.0)	84.7 (0.1)	90.5 (0.1)	87.5 (0.0)
	M3ID[23]	85.9 (4.4)	88.7 (0.4)	82.1 (10.4)	85.2 (6.1)	87.1 (0.4)	97.2 (0.4)	76.4 (0.9)	85.6 (0.4)	87.5 (0.4)	85.1 (0.5)	90.8 (0.2)	87.9 (0.4)
	RITUAL[74]	86.2 (4.7)	89.8 (1.5)	80.6 (8.9)	84.9 (5.8)	87.7 (0.2)	96.8 (0.0)	77.9 (0.6)	86.3 (0.3)	87.1 (0.0)	85.0 (0.4)	90.1 (0.5)	87.5 (0.0)
	DeGF[80]	86.7 (5.2)	90.0 (1.7)	82.4 (10.7)	86.0 (6.9)	87.9 (0.4)	96.7 (0.1)	78.6 (1.3)	86.7 (0.7)	86.9 (0.2)	83.9 (0.7)	90.2 (0.4)	86.9 (0.6)
	AGLA[3]	86.0 (4.5)	89.9 (1.6)	82.0 (10.3)	85.8 (6.7)	87.8 (0.3)	96.5 (0.3)	78.0 (0.7)	86.3 (0.3)	87.1 (0.0)	84.9 (0.3)	90.1 (0.5)	87.4 (0.1)
	EnAR	87.6 (6.1)	91.3 (3.0)	83.1 (11.4)	87.1 (8.0)	88.9 (1.4)	96.8 (0.0)	80.1 (2.8)	87.7 (1.7)	87.9 (0.8)	85.3 (0.7)	91.0 (0.4)	88.1 (0.6)
Adversarial	Regular	78.4	82.9	71.7	76.9	86.4	94.6	77.3	85.1	83.6	79.4	90.6	84.6
	VCD[39]	80.9 (2.5)	80.0 (2.9)	82.2 (10.5)	81.0 (4.1)	86.9 (0.5)	93.6 (1.0)	79.2 (1.9)	85.8 (0.7)	84.0 (0.4)	80.1 (0.7)	90.6 (0.0)	85.0 (0.4)
	M3ID[23]	81.4 (3.0)	82.1 (0.8)	82.2 (10.5)	82.1 (5.2)	86.1 (0.3)	94.7 (0.1)	76.4 (0.9)	84.6 (0.5)	84.1 (0.5)	80.0 (0.6)	90.8 (0.2)	85.0 (0.4)
	RITUAL[74]	81.8 (3.4)	82.4 (0.5)	80.6 (8.9)	81.5 (4.6)	86.6 (0.2)	94.0 (0.6)	77.8 (0.5)	85.1 (0.0)	83.7 (0.1)	79.8 (0.4)	90.2 (0.4)	84.7 (0.1)
	DeGF[80]	81.9 (3.5)	82.6 (0.3)	82.4 (10.7)	82.5 (5.6)	86.8 (0.4)	94.2 (0.4)	78.6 (1.3)	85.7 (0.6)	83.9 (0.3)	79.3 (0.1)	90.3 (0.3)	84.4 (0.2)
	AGLA[3]	82.2 (3.8)	82.2 (0.7)	82.0 (10.3)	82.1 (5.2)	86.5 (0.1)	94.4 (0.2)	78.0 (0.7)	85.5 (0.4)	83.8 (0.2)	80.1 (0.7)	90.1 (0.5)	84.8 (0.2)
	EnAR	83.9 (5.5)	84.5 (1.6)	83.1 (11.4)	83.8 (6.9)	87.5 (1.1)	94.5 (0.1)	80.1 (2.8)	86.8 (1.7)	84.2 (0.6)	80.3 (0.9)	90.9 (0.3)	85.3 (0.7)

increases from 71.35% to 75.62%, with overall WHOOPS accuracy rising from 49.34% to 61.32%. These results further demonstrate EnAR’s effectiveness and robustness in handling diverse counterfactual hallucination scenarios. Further details are presented in Appendix C.

HallusionBench. HallusionBench remains highly challenging, as it requires fine-grained reasoning over subtle inconsistencies between visual content and linguistic context. As shown in Table 2, for LLaVA-1.5-7B, EnAR achieves significant improvements of 4.0%, 1.6%, and 5.1% in aAcc, fAcc, and qAcc. Moreover, EnAR also enhances the performance of Qwen2.5VL-7B and InternVL3.5-8B, surpassing all competing methods. These results demonstrate that EnAR effectively mitigates both language hallucination and visual illusion problems in LVLMs.

POPE. Table 3 demonstrates EnAR’s performance on general hallucination benchmark. Across all three difficulty levels in POPE (Random, Popular, Adversarial) and three model architectures, our method delivers substantial gains and outperforms other methods. On LLaVA-1.5-7B, F1 score rises by 8.1%, 8.0%, and 6.9% for the three settings respectively. For Qwen2.5VL-7B and InternVL3.5-8B, our method achieves further improvements over the already strong baseline models. Notably, EnAR maintains balanced precision-recall trade-offs while achieving the best overall metrics. The consistent improvements confirm that our approach effectively mitigates object hallucination. EnAR utilizes the rich visual priors of diffusion models to synthesize contrastive visual inputs. As discussed in VCD [39], this can be viewed as injecting fine-grained contrastive noise into the original image input, achieving competitive performance on general hallucination tasks.

Table 4. **Ablation study on key components of our method across three LVLM backbones.** We report overall accuracy on VLMBias and average performance metrics on POPE benchmark.

Model	Setting	VLMBias		POPE	
		Overall	Acc.	Prec.	F1 Score
InternVL3.5-8B [71]	+EnAR	31.36	88.6	87.3	89.0
	w/o uncertainty map	29.03	88.2	86.9	88.9
	w/o visual impression	25.80	87.9	86.3	88.5
	w/o ours	19.83	87.6	86.1	88.2
Qwen2.5VL-7B [6]	+EnAR	28.02	88.8	96.6	87.8
	w/o uncertainty map	26.25	87.7	96.5	86.8
	w/o visual impression	24.50	87.4	96.5	86.1
	w/o ours	22.63	87.3	96.7	85.9
LLaVA-v1.5-7B [51]	+EnAR	22.20	87.1	90.5	86.6
	w/o uncertainty map	20.27	86.5	88.7	85.8
	w/o visual impression	17.78	84.3	88.2	83.4
	w/o ours	16.92	81.2	87.9	78.9

4.3. Ablation Study and Analysis

Key Component Ablation. We conduct ablation studies to evaluate the contributions of key components in our method, including the uncertainty map and visual impression. Without the visual impression, the counterfactual tokens are selected solely based on the uncertainty map derived from the original image input. As shown in Table 4, removing either component leads to notable performance drops across all three LVLM backbones on both the VLM-Bias and POPE benchmarks. The uncertainty map plays a crucial role in counterfactual element localization, while visual impression generation enhances the model’s understanding of visual scene. The full integration of both components yields the best results, confirming their complementary contributions to reducing hallucination. The detailed results are provided in Appendix D.1. Detailed analysis of EnAR’s efficiency and potential solutions to reduce

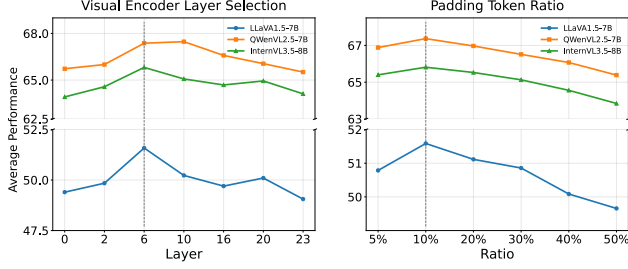


Figure 4. Average performance variation across VLM biases, POPE, and HallusionBench under different settings: (left) selection of vision encoder layer, (right) padding token ratio.

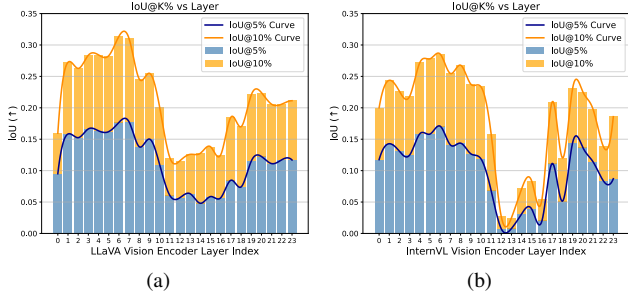


Figure 5. IoU between top-K% tokens from attention maps at different encoder layers and salient object regions: (a) LLaVA-1.5-7B, (b) InternVL3.5-8B.

inference cost are provided in Appendix E.

Encoder Layer Selection. We analyze the impact of selecting different encoder layers for extracting the attention map. As shown in Figure 4 (left), selecting layers around the 6th layer yields better average performance, outperforming the other layers. To further investigate how layer depth influences performance, we employ the HKU-IS dataset [41], a dataset for salient object detection. As illustrated in Figure 5, within the vision encoder of LLaVA-v1.5-7B and InternVL3.5-8B, we identified the image regions covered by top-K% tokens from attention maps at different layers and computed their IoU with salient object regions on HKU-IS. When averaged over 4,447 samples, the shallow layers achieve higher coverage. Consistent with prior observations [17, 33, 46, 81], we attribute this to early layers focusing more on concrete object features, while mid-to-deep layers exhibit attention drift, where tokens at edge and background positions act like registers [17] that aggregate complex global visual information. Since EnAR requires attending to tokens located at counterfactual elements, lower layers are more suitable. Therefore, we simply adopt the 6th layer across all baseline models. Appendices D.2 and D.4 provide detailed results and visualizations.

Padding Token Ratio Ablation. We investigate the effect of different padding token ratios on model performance. As shown in Figure 4 (right), setting the padding token ratio at 10% yields the best results. We argue that too few

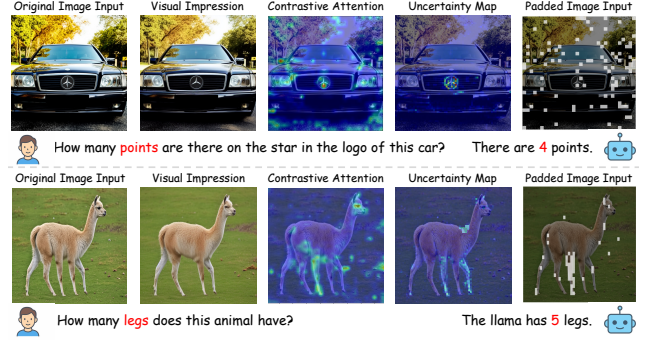


Figure 6. **Case study visualization.** We illustrate how EnAR constructs visual impressions, localizes counterfactual elements, and produces corrected responses.

padding tokens may fail to sufficiently cover counterfactual elements, while too many can cause information dispersion and hinder focus. Therefore, we simply adopt a 10% padding token ratio for all baseline models. Further details can be found in Appendix D.3.

4.4. Case Study

We present qualitative examples to illustrate the characteristics of our framework in Figure 6. For each original image input, we first construct its corresponding visual impression (e.g., transforming a five-legged llama into a normal four-legged one). EnAR then feeds both images into the LVLm’s vision encoder to obtain a contrastive attention map, which clearly highlights the extra leg. In combination with the uncertainty map, EnAR further identifies these regions as counterfactual tokens. Then EnAR replaces them with the $\langle pad \rangle$ token to form the padded input. Finally, contrastive decoding between the original and padded inputs rectifies the model’s response. See Appendix F for more cases.

5. Conclusion

Inspired by human cognition, we propose the Envision-Attend-Respond (EnAR) framework to tackle the challenging problem of counterfactual hallucination in LVLms. EnAR first constructs a visual impression under real-world visual priors, then attends to counterfactual tokens by contrasting attentions between the impression and the original image, further guided by uncertainty cues to pad hallucination-prone regions. Finally, it performs contrastive decoding between the original and padded visual inputs. In this way, EnAR effectively alleviates both counterfactual and general hallucinations without any additional training. Our findings suggest that leveraging visual impressions as a reliable reference offers a promising direction for hallucination governance in LVLms, and future work can build on this foundation to design more fine-grained mechanisms for grounding multimodal understanding.

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References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 2
- [2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 1
- [3] Wenbin An, Feng Tian, Sicong Leng, Jiahao Nie, Haonan Lin, QianYing Wang, Ping Chen, Xiaoqin Zhang, and Shijian Lu. Mitigating object hallucinations in large vision-language models with assembly of global and local attention. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 29915–29926, 2025. 2, 3, 5, 6, 7
- [4] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023. 1
- [5] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 1(2):3, 2023. 1
- [6] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 2, 5, 6, 7, 3
- [7] Nitzan Bitton-Guetta, Yonatan Bitton, Jack Hessel, Ludwig Schmidt, Yuval Elovici, Gabriel Stanovsky, and Roy Schwartz. Breaking common sense: Whoops! a vision-and-language benchmark of synthetic and compositional images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2616–2627, 2023. 1, 3, 6
- [8] Chi Chen, Ruoyu Qin, Fuwen Luo, Xiaoyue Mi, Peng Li, Maosong Sun, and Yang Liu. Position-enhanced visual instruction tuning for multimodal large language models. *arXiv preprint arXiv:2308.13437*, 2023. 2, 3
- [9] Jun Chen, Deyao Zhu, Xiaoqian Shen, Xiang Li, Zechun Liu, Pengchuan Zhang, Raghuraman Krishnamoorthi, Vikas Chandra, Yunyang Xiong, and Mohamed Elhoseiny. Minigt-v2: large language model as a unified interface for vision-language multi-task learning. *arXiv preprint arXiv:2310.09478*, 2023. 1
- [10] Zhiyang Chen, Yousong Zhu, Yufei Zhan, Zhaowen Li, Chaoyang Zhao, Jinqiao Wang, and Ming Tang. Mitigating hallucination in visual language models with visual supervision. *arXiv preprint arXiv:2311.16479*, 2023. 1
- [11] Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, et al. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *Science China Information Sciences*, 67(12):220101, 2024. 2
- [12] Tushar Choudhary, Vikrant Dewangan, Shivam Chandhok, Shubham Priyadarshan, Anushka Jain, Arun K Singh, Siddharth Srivastava, Krishna Murthy Jatavallabhula, and K Madhava Krishna. Talk2bev: Language-enhanced bird’s-eye view maps for autonomous driving. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 16345–16352. IEEE, 2024. 1
- [13] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113, 2023. 1
- [14] Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and brain sciences*, 36(3):181–204, 2013. 2
- [15] Wenliang Dai, Junnan Li, DONGXU LI, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. In *NeurIPS*, pages 49250–49267, 2023. 1, 2
- [16] Wenliang Dai, Zihan Liu, Ziwei Ji, Dan Su, and Pascale Fung. Plausible may not be faithful: Probing object hallucination in vision-language pre-training. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2136–2148, 2023. 2
- [17] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *The Twelfth International Conference on Learning Representations*, 2024. 8, 5, 6
- [18] Floris P De Lange, Micha Heilbron, and Peter Kok. How do expectations shape perception? *Trends in cognitive sciences*, 22(9):764–779, 2018. 2
- [19] Hanneke EM Den Ouden, Peter Kok, and Floris P De Lange. How prediction errors shape perception, attention, and motivation. *Frontiers in psychology*, 3:548, 2012. 2
- [20] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 2
- [21] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. 1
- [22] Bradley Efron. Tweedie’s formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614, 2011. 4
- [23] Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille,

- Ashwin Swaminathan, and Stefano Soatto. Multi-modal hallucination control by visual information grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14303–14312, 2024. [2](#), [3](#), [5](#), [6](#), [7](#)
- [24] Karl Friston. The free-energy principle: a unified brain theory? *Nature reviews neuroscience*, 11(2):127–138, 2010. [2](#)
- [25] Matthias Fritsche, Eelke Spaak, and Floris P De Lange. A bayesian and efficient observer model explains concurrent attractive and repulsive history biases in visual perception. *Elife*, 9:e55389, 2020. [2](#)
- [26] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, Rongrong Ji, Caifeng Shan, and Ran He. Mme: A comprehensive evaluation benchmark for multi-modal large language models, 2025. [3](#)
- [27] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014. [4](#)
- [28] Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, et al. Hallusionbench: an advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *CVPR*, pages 14375–14385, 2024. [1](#), [3](#), [6](#), [2](#)
- [29] Anisha Gunjal, Jihan Yin, and Erhan Bas. Detecting and preventing hallucinations in large vision language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18135–18143, 2024. [2](#)
- [30] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. [2](#), [4](#)
- [31] Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang, Conghui He, Jiaqi Wang, Dahua Lin, Weiming Zhang, and Nenghai Yu. Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13418–13427, 2024. [2](#)
- [32] Jiefeng Jiang, Christopher Summerfield, and Tobias Egner. Attention sharpens the distinction between expected and unexpected percepts in the visual brain. *Journal of Neuroscience*, 33(47):18438–18447, 2013. [2](#)
- [33] Seil Kang, Jinyeong Kim, Junhyeok Kim, and Seong Jae Hwang. See what you are told: Visual attention sink in large multimodal models. In *The Thirteenth International Conference on Learning Representations*, 2025. [8](#), [5](#)
- [34] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *International Conference on Learning Representations*, 2014. [4](#)
- [35] Peter Kok, Janneke FM Jehee, and Floris P De Lange. Less is more: expectation sharpens representations in the primary visual cortex. *Neuron*, 75(2):265–270, 2012. [2](#)
- [36] Peter Kok, Michel F Failing, and Floris P de Lange. Prior expectations evoke stimulus templates in the primary visual cortex. *Journal of cognitive neuroscience*, 26(7):1546–1554, 2014. [2](#)
- [37] Peter Kok, Pim Mostert, and Floris P De Lange. Prior expectations induce prestimulus sensory templates. *Proceedings of the National Academy of Sciences*, 114(39):10473–10478, 2017. [2](#)
- [38] Hugo Laurençon, Lucile Saulnier, Leo Tronchon, Stas Bekman, Amanpreet Singh, Anton Lozhkov, Thomas Wang, Siddharth Karamcheti, Alexander Rush, Douwe Kiela, Matthieu Cord, and Victor Sanh. Obelics: An open web-scale filtered dataset of interleaved image-text documents. In *Advances in Neural Information Processing Systems*, pages 71683–71702. Curran Associates, Inc., 2023. [1](#)
- [39] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13872–13882, 2024. [2](#), [3](#), [5](#), [6](#), [7](#)
- [40] Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36, 2024. [1](#)
- [41] Guanbin Li and Yizhou Yu. Visual saliency based on multi-scale deep features. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5455–5463, 2015. [8](#), [5](#)
- [42] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023. [1](#)
- [43] Xu Li, Yi Zheng, Haotian Chen, Xiaolei Chen, Yuxuan Liang, Chenghang Lai, Bin Li, and Xiangyang Xue. Instruction-guided fusion of multi-layer visual features in large vision-language models. *Pattern Recognition*, 170: 111932, 2026. [1](#)
- [44] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. [1](#), [2](#), [3](#), [6](#)
- [45] Zhang Li, Biao Yang, Qiang Liu, Zhiyin Ma, Shuo Zhang, Jingxu Yang, Yabo Sun, Yuliang Liu, and Xiang Bai. Monkey: Image resolution and text label are important things for large multi-modal models. In *CVPR*, pages 26763–26773, 2024. [2](#)
- [46] Yuxuan Liang, Xu Li, Xiaolei Chen, Haotian Chen, Yi Zheng, Chenghang Lai, Bin Li, and Xiangyang Xue. Global semantic-guided sub-image feature weight allocation in high-resolution large vision-language models. *arXiv preprint arXiv:2501.14276*, 2025. [2](#), [8](#), [5](#)
- [47] Yuxuan Liang, Xu Li, Xiaolei Chen, Haotian Chen, Yi Zhen, Zhe Liu, Rui Zhu, Bin Li, and Xiangyang Xue. Pyramid token pruning for high-resolution large vision-language models via region, token, and instruction-guided importance. *IEEE Transactions on Circuits and Systems for Video Technology*, pages 1–1, 2026. [1](#)
- [48] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence

- Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pages 740–755. Springer, 2014. 6
- [49] Fuxiao Liu, Kevin Lin, Linjie Li, Jianfeng Wang, Yaser Yacoob, and Lijuan Wang. Mitigating hallucination in large multi-modal models via robust instruction tuning. *arXiv preprint arXiv:2306.14565*, 2023. 1
- [50] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, pages 34892–34916, 2023. 1
- [51] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26296–26306, 2024. 1, 2, 5, 6, 7, 3
- [52] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, 2024. 2
- [53] Jiazhen Liu, Yuhan Fu, Ruobing Xie, Runquan Xie, Xingwu Sun, Fengzong Lian, Zhanhui Kang, and Xirong Li. Phd: A chatgpt-prompted visual hallucination evaluation dataset. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19857–19866, 2025. 1, 3, 6
- [54] Junyu Lu, Dixiang Zhang, Songxin Zhang, Zejian Xie, Zhuoyang Song, Cong Lin, Jiaying Zhang, Bingyi Jing, and Pingjian Zhang. Lyrics: Boosting fine-grained language-vision alignment and comprehension via semantic-aware visual objects. *arXiv preprint arXiv:2312.05278*, 2023. 2, 3
- [55] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sd-xl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023. 2
- [56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmlR, 2021. 2
- [57] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 2, 4, 6, 1
- [58] Antonio Sclocchi, Alessandro Favero, and Matthieu Wyart. A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 122(1):e2408799121, 2025. 4
- [59] Fan Shi, Bin Li, and Xiangyang Xue. Towards generative abstract reasoning: Completing raven’s progressive matrix via rule abstraction and selection. In *The Twelfth International Conference on Learning Representations*, 2024. 4
- [60] Fan Shi, Bin Li, and Xiangyang Xue. Abstracting concept-changing rules for solving raven’s progressive matrix problems. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–12, 2026. 4
- [61] Fan Shi, Yuxuan Liang, Xiaolei Chen, Haiyang Yu, Xu Li, Yi Zheng, Rui Zhu, Xiangyang Xue, and Bin Li. Decomposition of concept-level rules in visual scenes. In *The Fourteenth International Conference on Learning Representations*, 2026. 1
- [62] Alyssa H Sinclair, Grace M Manalili, Iva K Brunec, R Alison Adcock, and Morgan D Barense. Prediction errors disrupt hippocampal representations and update episodic memories. *Proceedings of the National Academy of Sciences*, 118(51):e2117625118, 2021. 2
- [63] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *International Conference on Learning Representations*, 2021. 2, 4
- [64] Christopher Summerfield and Floris P De Lange. Expectation in perceptual decision making: neural and computational mechanisms. *Nature Reviews Neuroscience*, 15(11):745–756, 2014. 2
- [65] Christopher Summerfield and Tobias Egner. Expectation (and attention) in visual cognition. *Trends in cognitive sciences*, 13(9):403–409, 2009. 2
- [66] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? exploring the visual shortcomings of multimodal llms. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9568–9578, 2024. 2, 3
- [67] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023. 1
- [68] An Vo, Khai-Nguyen Nguyen, Mohammad Reza Tarsi, Vy Tuong Dang, Anh Totti Nguyen, and Daeyoung Kim. Vision language models are biased. *arXiv preprint arXiv:2505.23941*, 2025. 1, 2, 3, 6
- [69] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024. 2
- [70] Weihang Wang, Qingsong Lv, Wenmeng Yu, Wenyi Hong, Ji Qi, Yan Wang, Junhui Ji, Zhuoyi Yang, Lei Zhao, Song XiXuan, et al. Cogvlm: Visual expert for pretrained language models. *Advances in Neural Information Processing Systems*, 37:121475–121499, 2024. 2
- [71] Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025. 2, 5, 6, 7, 3
- [72] Haoran Wei, Lingyu Kong, Jinyue Chen, Liang Zhao, Zheng Ge, Jinrong Yang, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Vary: Scaling up the vision vocabulary for large vision-language model. In *European Conference on Computer Vision*, pages 408–424. Springer, 2024. 2, 3
- [73] Max Welling and Yee W Teh. Bayesian learning via stochastic gradient langevin dynamics. In *Proceedings of the 28th international conference on machine learning (ICML-11)*, pages 681–688, 2011. 4
- [74] Sangmin Woo, Jaehyuk Jang, Donguk Kim, Yubin Choi, and Changick Kim. Ritual: Random image transformations as

- a universal anti-hallucination lever in large vision language models, 2024. [2](#), [3](#), [5](#), [6](#), [7](#)
- [75] Yuanchen Wu, Lu Zhang, Hang Yao, Junlong Du, Ke Yan, Shouhong Ding, Yunsheng Wu, and Xiaoqiang Li. Antidote: A unified framework for mitigating lvlm hallucinations in counterfactual presupposition and object perception. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 14646–14656, 2025. [2](#), [3](#)
 - [76] Zhiyu Wu, Xiaokang Chen, Zizheng Pan, Xingchao Liu, Wen Liu, Damai Dai, Huazuo Gao, Yiyang Ma, Chengyue Wu, Bingxuan Wang, et al. Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding. *arXiv preprint arXiv:2412.10302*, 2024. [2](#)
 - [77] Qinghao Ye, Haiyang Xu, Jiabo Ye, Ming Yan, Anwen Hu, Haowei Liu, Qi Qian, Ji Zhang, and Fei Huang. mplug-owl2: Revolutionizing multi-modal large language model with modality collaboration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13040–13051, 2024. [2](#)
 - [78] Qifan Yu, Juncheng Li, Longhui Wei, Liang Pang, Wentao Ye, Bosheng Qin, Siliang Tang, Qi Tian, and Yueting Zhuang. Hallucidoctor: Mitigating hallucinatory toxicity in visual instruction data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12944–12953, 2024. [1](#), [3](#)
 - [79] Tianyu Yu, Yuan Yao, Haoye Zhang, Taiwen He, Yifeng Han, Ganqu Cui, Jinyi Hu, Zhiyuan Liu, Hai-Tao Zheng, Maosong Sun, et al. Rlhv-v: Towards trustworthy mllms via behavior alignment from fine-grained correctional human feedback. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13807–13816, 2024. [1](#), [2](#), [3](#)
 - [80] Ce Zhang, Zifu Wan, Zhehan Kan, Martin Q. Ma, Simon Stepputtis, Deva Ramanan, Russ Salakhutdinov, Louis-Philippe Morency, Katia P. Sycara, and Yaqi Xie. Self-correcting decoding with generative feedback for mitigating hallucinations in large vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025. [2](#), [3](#), [5](#), [6](#), [7](#)
 - [81] Qizhe Zhang, Aosong Cheng, Ming Lu, Renrui Zhang, Zhiyong Zhuo, Jiajun Cao, Shaobo Guo, Qi She, and Shanghang Zhang. Beyond text-visual attention: Exploiting visual cues for effective token pruning in vlms. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20857–20867, 2025. [8](#), [5](#)
 - [82] Zefeng Zhang, Hengzhu Tang, Jiawei Sheng, Zhenyu Zhang, Yiming Ren, Zhenyang Li, Dawei Yin, Duohe Ma, and Tingwen Liu. Debiasing multimodal large language models via noise-aware preference optimization. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 9423–9433, 2025. [2](#), [3](#)
 - [83] Deyao Zhu, jun chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigt-4: Enhancing vision-language understanding with advanced large language models. In *International Conference on Representation Learning*, pages 18378–18394, 2024. [2](#)